

Gromov's tori are optimal

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Abstract

We give an optimal bound on normal curvatures of immersed n -torus in a Euclidean ball of large dimension.

1 Introduction

Let us denote by $\mathbb{B}^q$ the closed unit ball in $\mathbb{R}^q$ centered at the origin. Further, $\mathbb{T}^n$ will denote the n -dimensional torus — the smooth manifold diffeomorphic to the product of n circles.

This note is inspired by examples of embeddings $\mathbb{T}^n \hookrightarrow \mathbb{B}^q$ for large q with constant normal curvatures $K_n = \sqrt{3 \cdot n / (n + 2)}$. In other words, any geodesic in the torus has constant curvature K_n as a curve in $\mathbb{R}^q$. These examples were found by Michael Gromov among geodesic subtori in Clifford's tori [5, 2.A], [4, 1.1.A.]. In particular, Gromov's tori have flat induced metrics. (Recall that Clifford's torus is a product of m circles of radius $1/\sqrt{m}$ in $\mathbb{R}^{2 \cdot m}$; its normal curvatures lie in the range $[1, \sqrt{m}]$.)

Gromov's examples lead to the following surprising facts: *any closed smooth manifold L admits a smooth embedding into $\mathbb{B}^q$ for large q with normal curvatures less than $\sqrt{3}$; moreover, the induced Riemannian metric on L can be chosen to be proportional to any given metric g ; see [5, 1.D] and [4, 1.1.C].*

The next theorem implies that Gromov's tori have the best upper bound on normal curvatures; in particular, the $\sqrt{3}$ -bound is optimal.

1.1. Theorem. *Suppose $\mathbb{T}^n$ is smoothly immersed in $\mathbb{B}^q$. Then its maximal normal curvature is at least*

$$\sqrt{3 \cdot \frac{n}{n+2}}.$$

To make the statement more exact, we need one more notation. Assume that L is a smooth n -dimensional manifold immersed in $\mathbb{R}^q$; we will always assume that L is equipped with the induced Riemannian metric. Let us denote by T_x and N_x the tangent and normal spaces of L at x .

Recall that the *second fundamental form* $\mathbb{I}$ at x is a symmetric quadratic form on T_x with values in N_x . It is uniquely defined by the identity $\mathbb{I}(v, v) \equiv \gamma_v''(0)$, where $v \in T_x$ and γ_v is an L -geodesic that starts at x with initial velocity vector v .

Given $x \in L$, denote by $\mathcal{K}(x)$ the average value of $|\mathbb{I}(u, u)|^2$ for $u \in T_x$ such that $|u| = 1$. Since $K(u) = |\mathbb{I}(u, u)|$ is the normal curvature in the direction u , we have that $\mathcal{K}(x)$ is the average value $K^2(u)$. (The Cyrillic zhe $\mathcal{K}$ is used since it resembles K^2 .)

1.2. Theorem. *Suppose $\mathbb{T}^n$ is smoothly immersed in $\mathbb{B}^q$. Let us equip $\mathbb{T}^n$ with the induced Riemannian metric; so we can take average values with respect to the induced volume.*

- (a) *If $n = 2$, then the average value of $\mathcal{K}$ is at least $\frac{3}{2}$.*
- (b) *If the metric on $\mathbb{T}^n$ is flat, then the average value of $\mathcal{K}$ is at least $3 \cdot \frac{n}{n+2}$.*
- (c) *If the image of $\mathbb{T}^n$ lies in $\partial\mathbb{B}^q$, then $\mathcal{K} \geq 3 \cdot \frac{n}{n+2}$ at some point of $\mathbb{T}^n$.*
- (d) *If $n \leq 4$, then $\mathcal{K} \geq 3 \cdot \frac{n}{n+2}$ at some point of $\mathbb{T}^n$.*
- (e) *If the normal curvatures of $\mathbb{T}^n$ do not exceed 2, then $\mathcal{K} \geq 3 \cdot \frac{n}{n+2}$ at some point of $\mathbb{T}^n$.*

Note that part (e) is a stronger version of 1.1. The remaining statements (a)–(d) are stronger versions of 1.1 in some partial cases. All this follows since the normal curvature in some direction at x is at least $\sqrt{\mathcal{K}(x)}$.

All proofs use our version of the Gauss formula; see below. The proofs of (c)–(e) use in addition that *the torus does not admit a metric with positive scalar curvature* [7, Corollary A].

1.3. Open question. *Is it true that for any smooth immersion $\mathbb{T}^n \looparrowright \mathbb{B}^q$ the inequality $\mathcal{K} \geq 3 \cdot \frac{n}{n+2}$ holds at some point?*

1.4. Open question. *Suppose $\mathbb{RP}^n \looparrowright \mathbb{B}^q$ is a smooth immersion. Is it true that its normal curvature is at least $\sqrt{2 \cdot n / (n+1)}$ in some direction?*

The last question asks if *the Veronese embedding is optimal*. Recall that the Veronese embedding $\mathbb{RP}^n \hookrightarrow \mathbb{B}^q$ has all normal curvatures $\sqrt{2 \cdot n / (n+1)}$; here $q = \frac{1}{2} \cdot (n+1) \cdot (n+2)$. An analogous question for immersions into unit spheres is open as well [9]. You may also ask for the optimal constant for your favorite closed manifold.

Gromov's construction gives many smooth immersions $\mathbb{T}^n \looparrowright \mathbb{B}^q$ with normal curvatures $\sqrt{3 \cdot n / (n+2)}$ in all directions. The proof of our theorem implies that such immersions send torus to the boundary of $\mathbb{B}^q$ and it tell exactly what is its second fundamenal form. Nevertheless the following question is open.

1.5. Open question. *Can one describe all smooth immersions $\mathbb{T}^n \looparrowright \mathbb{B}^q$ with normal curvatures $\sqrt{3 \cdot n / (n+2)}$ in all directions?*

2 Gauss formula

Recall that L is a smooth n -dimensional manifold immersed in $\mathbb{R}^q$. Given $p \in L$, denote by $\text{Sc}(p)$ and $H(p)$ the scalar curvature and the mean curvature vector at p .

The following version of the Gauss formula plays a central role in all proofs; it is used instead of the formula in [6, 5.B].

2.1. Gauss formula. *The following identity*

$$\text{Sc} = \frac{3}{2} \cdot |H|^2 - \frac{n \cdot (n+2)}{2} \cdot \mathcal{K}$$

holds for any smooth n -dimensional immersed manifold in a Euclidean space.

Proof. Choose a point $p \in L$.

Assume $\text{codim } L = 1$. Denote by $k_1, \dots, k_n$ the principal curvatures of L at p . Note that

$$|H|^2 = \sum_i k_i^2 + 2 \cdot \sum_{i < j} k_i \cdot k_j.$$

Further,

$$n \cdot (n+2) \cdot \mathcal{K} = 3 \cdot \sum_i k_i^2 + 2 \cdot \sum_{i < j} k_i \cdot k_j.$$

The last identity follows since $\mathcal{K}$ is the average value of $(\sum_i k_i \cdot x_i^2)^2$ on the unit sphere $\mathbb{S}^{n-1} \subset \mathbb{R}^n = T_p$; here $(x_1, \dots, x_n)$ are the standard coordinates in $\mathbb{R}^n$. One has to take into account that the following functions have unit average values: $\frac{1}{3} \cdot n \cdot (n+2) \cdot x_i^4$ and $n \cdot (n+2) \cdot x_i^2 \cdot x_j^2$ for $i \neq j$.

By the standard Gauss formula,

$$\text{Sc} = 2 \cdot \sum_{i < j} k_i \cdot k_j.$$

It remains to rewrite the right-hand side using the expressions for $|H|^2$ and $\mathcal{K}$.

If $\text{codim } L = k > 1$, then the second fundamental form at p can be presented as a direct sum of k real-valued quadratic forms $\mathbb{I}_1 \oplus \dots \oplus \mathbb{I}_k$; that is,

$$\mathbb{I} = e_1 \cdot \mathbb{I}_1 + \dots + e_k \cdot \mathbb{I}_k,$$

where $e_1, \dots, e_k$ is an orthonormal basis of N_p . Denote by Sc_i , H_i , and $\mathcal{K}_i$ the values associated with $\mathbb{I}_i$. From above, we get

$$\text{Sc}_i = \frac{3}{2} \cdot |H_i|^2 - \frac{n \cdot (n+2)}{2} \cdot \mathcal{K}_i$$

for each i .

Note that

$$\text{Sc} = \sum_i \text{Sc}_i, \quad |H|^2 = \sum_i |H_i|^2, \quad \text{and} \quad \mathcal{K} = \sum_i \mathcal{K}_i.$$

Hence the general case follows. $\square$

Remark. A more direct proof of this formula can be obtained using the so-called *extrinsic curvature tensor* which is defined by $\Phi(x, y, v, w) := \langle \mathbb{I}(x, y), \mathbb{I}(v, w) \rangle$; the necessary properties of this tensor are discussed in [8]. As a bonus, one gets an explicit expression for the second fundamental forms of all Gromov's tori.

3 Special cases

The following statement appears in the book of Yuri Burago and Viktor Zalgaller [2, Theorem 28.2.5]; it generalizes the result of István Fáry about average curvature of a curve in the unit ball [3, 12], but the proof is essentially the same.

3.1. Lemma. *Let L be a closed n -dimensional manifold that is smoothly immersed in $\mathbb{B}^q$. Then the average value of $|H|$ on L is at least n .*

Proof. Consider the function $u: x \mapsto \frac{1}{2} \cdot |x|^2$ on L . Note that

$$(\Delta u)(x) = n + \langle H(x), x \rangle.$$

It follows that the average value of $\langle H(x), x \rangle$ is $-n$. Since $|x| \leq 1$, we get the result. $\square$

Proof of 1.2a. By 3.1, the average value of $|H|^2$ is at least 4. Further, by the Gauss–Bonnet formula, the scalar curvature (which is twice the Gauss curvature in this case) has zero average. Therefore 2.1 implies the statement. $\square$

Proof of 1.2b. By 3.1, the average value of $|H|^2$ is at least n^2 . Since $\text{Sc} \equiv 0$, it remains to apply 2.1. $\square$

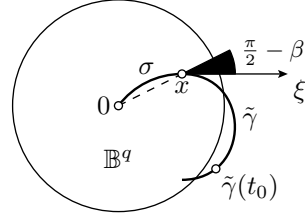
Proof of 1.2c. Since the image lies in the unit sphere, we have that $|H|^2$ is at least n^2 at each point. Since $\mathbb{T}^n$ does not admit a metric with positive scalar curvature [7, Corollary A], we have $\text{Sc}(x) \leq 0$ at some point x . It remains to apply 2.1 at x . $\square$

4 Main case

The following lemma is an easy corollary of the bow lemma of Axel Schur [10, 11]. It explains how we use the assumption on normal curvatures in 1.2e. If $n \leq 4$, then the proof works without this assumption.

4.1. Lemma. *Let L be a complete smooth manifold smoothly immersed in $\mathbb{B}^q$. Suppose its normal curvatures are at most 2. Given $x \in L$, denote by $\beta = \beta(x)$ the angle between vector x and the normal space N_x . Then $|x| \leq \cos \beta$.*

Proof. Let ξ be a tangent direction at x such that $\angle(x, \xi) = \frac{\pi}{2} - \beta$. In the plane spanned by x and ξ , choose a unit-speed circle arc σ from 0 to x that comes to x in the direction opposite to ξ ; extend σ after x by a unit-speed semicircle $\tilde{\gamma}$ with curvature 2 in such a way that the concatenation $\sigma * \tilde{\gamma}$ is an arc of a C^1 -smooth convex plane curve; see the figure.



Observe that if $|x| > \cos \beta$, then $\tilde{\gamma}$ leaves $\mathbb{B}^q$; that is, $|\tilde{\gamma}(t_0)| > 1$ for some t_0 .

Let γ be the unit-speed geodesic in L that runs from x in the direction ξ . Note that curvatures of $\sigma * \gamma$ do not exceed the curvatures of $\sigma * \tilde{\gamma}$ at the corresponding points. Applying the bow lemma for $\sigma * \gamma$ and $\sigma * \tilde{\gamma}$, we get $|\gamma(t_0)| \geq |\tilde{\gamma}(t_0)|$. It follows that L does not lie in $\mathbb{B}^q$ — a contradiction. $\square$

Let g be a Riemannian metric on $\mathbb{T}^n$. Suppose $n \geq 3$, and $u: \mathbb{T}^n \rightarrow \mathbb{R}$ is a smooth positive function. Recall that

$$\left(\text{Sc} \cdot u - 4 \cdot \frac{n-1}{n-2} \cdot \Delta u \right) \cdot u^{-\frac{n+2}{n-2}}$$

is the scalar curvature of the metric $u^{\frac{4}{n-2}} \cdot g$; see for example [1, 6.3]. Since any Riemannian metric on $\mathbb{T}^n$ has nonpositive scalar curvature at some point [7, Corollary A], we get the following.

4.2. Claim. *For any Riemannian metric on $\mathbb{T}^n$ and any positive smooth function $u: \mathbb{T}^n \rightarrow \mathbb{R}$, the function*

$$\text{Sc} \cdot u - 4 \cdot \frac{n-1}{n-2} \cdot \Delta u$$

returns a nonpositive value at some point.

Proof of 1.2d and 1.2e. The case $n = 2$ follows from 1.2a; so we can assume that $n \geq 3$. Consider the function $u: x \mapsto \exp(-\frac{k}{2} \cdot |x|^2)$ for $x \in \mathbb{T}^n$.

We will apply the following formula

$$\Delta(\varphi \circ f) = (\varphi' \circ f) \cdot \Delta f + (\varphi'' \circ f) \cdot |\nabla f|^2$$

to $f: x \mapsto \frac{1}{2} \cdot |x|^2$ and $\varphi: y \mapsto \exp(-k \cdot y)$; so $u = \varphi \circ f$.

Set $r(x) = |x|$, $\alpha(x) = \angle(H(x), x)$, and $\beta(x)$ as in 4.1. Note that

$$\textcircled{1} \quad \beta \leq \alpha \leq \pi - \beta.$$

Observe that

$$\Delta f = |H| \cdot r \cdot \cos \alpha + n, \quad |\nabla f| = r \cdot \sin \beta, \quad \varphi' = -k \cdot \varphi, \quad \varphi'' = k^2 \cdot \varphi.$$

Therefore

$$\Delta u = u \cdot [-k \cdot |H| \cdot r \cdot \cos \alpha - k \cdot n + k^2 \cdot r^2 \cdot \sin^2 \beta].$$

Recall that $\text{Sc} = -\frac{n \cdot (n+2)}{2} \cdot \mathcal{K} + \frac{3}{2} \cdot |H|^2$; see 2.1. By 4.2, the function

$$\begin{aligned} \text{Sc} \cdot u - 4 \cdot \frac{n-1}{n-2} \cdot \Delta u = u \cdot \left[-\frac{n \cdot (n+2)}{2} \cdot \mathcal{K} + \frac{3}{2} \cdot |H|^2 + 4 \cdot \frac{n-1}{n-2} \cdot k \cdot |H| \cdot r \cdot \cos \alpha + \right. \\ \left. + 4 \cdot \frac{n-1}{n-2} \cdot (k \cdot n - k^2 \cdot r^2 \cdot \sin^2 \beta) \right] \end{aligned}$$

returns a nonpositive value at some point $x \in \mathbb{T}^n$.

Choose

$$k = \frac{3}{4} \cdot \frac{n-2}{n-1} \cdot n, \quad \text{so} \quad n = \frac{4}{3} \cdot \frac{n-1}{n-2} \cdot k.$$

At the point x , we have

$$\textcircled{2} \quad \begin{aligned} \frac{n \cdot (n+2)}{2} \cdot \mathcal{K} \geq \frac{3}{2} \cdot (|H| + n \cdot r \cdot \cos \alpha)^2 - \frac{3}{2} \cdot n^2 \cdot r^2 \cdot \cos^2 \alpha + \\ + 3 \cdot n^2 - \frac{9}{4} \cdot \frac{n-2}{n-1} \cdot n^2 \cdot r^2 \cdot \sin^2 \beta \geq \frac{3}{2} \cdot n^2. \end{aligned}$$

Indeed, by $\textcircled{1}$, $\cos^2 \alpha + \sin^2 \beta \leq 1$. If $n \leq 4$, then $\frac{3}{2} \geq \frac{9}{4} \cdot \frac{n-2}{n-1}$; therefore the last inequality follows, and it proves 1.2d.

Further, if $n \geq 5$, then for the last inequality in $\textcircled{2}$ we need to use in addition that $r^2 + \sin^2 \beta \leq 1$ which follows from 4.1. Hence 1.2e follows. $\square$

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References

- [1] T. Aubin. *Nonlinear analysis on manifolds. Monge–Ampère equations*. Vol. 252. Grundlehren der mathematischen Wissenschaften. 1982.

- [2] Yu. D. Burago and V. A. Zalgaller. *Geometric inequalities*. Vol. 285. Grundlehren der mathematischen Wissenschaften. 1988.
- [3] I. Fáry. “Sur certaines inégalités géométriques”. *Acta Sci. Math.* 12 (1950), 117–124.
- [4] M. Gromov. *Curvature, Kolmogorov diameter, Hilbert rational designs and over-twisted immersions*. 2022. arXiv: 2210.13256 [math.DG].
- [5] M. Gromov. *Isometric immersions with controlled curvatures*. 2022. arXiv: 2212.06122v1 [math.DG].
- [6] M. Gromov. *Scalar curvature, injectivity radius and immersions with small second fundamental forms*. 2022. arXiv: 2203.14013v2 [math.DG].
- [7] M. Gromov and B. Lawson. “Spin and scalar curvature in the presence of a fundamental group. I”. *Ann. of Math. (2)* 111.2 (1980), 209–230.
- [8] A. Petrunin. “Polyhedral approximations of Riemannian manifolds”. *Turkish J. Math.* 27.1 (2003), 173–187.
- [9] A. Petrunin. *Normal curvature of Veronese embedding*. MathOverflow. eprint: <https://mathoverflow.net/q/445819>.
- [10] A. Petrunin and S. Zamora Barrera. *What is differential geometry: curves and surfaces*. 2022. arXiv: 2012.11814 [math.HO].
- [11] A. Schur. “Über die Schwarzsche Extremaleigenschaft des Kreises unter den Kurven konstanter Krümmung”. *Math. Ann.* 83.1-2 (1921), 143–148.
- [12] S. Tabachnikov. “The tale of a geometric inequality.” *MASS selecta: teaching and learning advanced undergraduate mathematics*. 2003, 257–262.